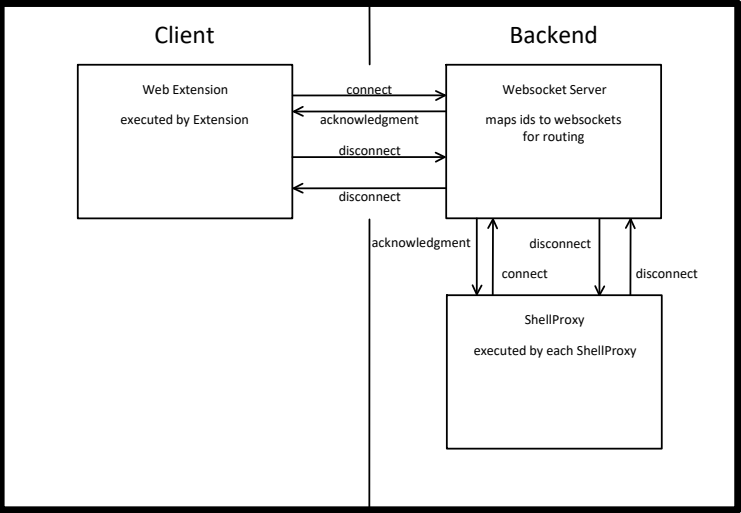


Connect & Disconnect



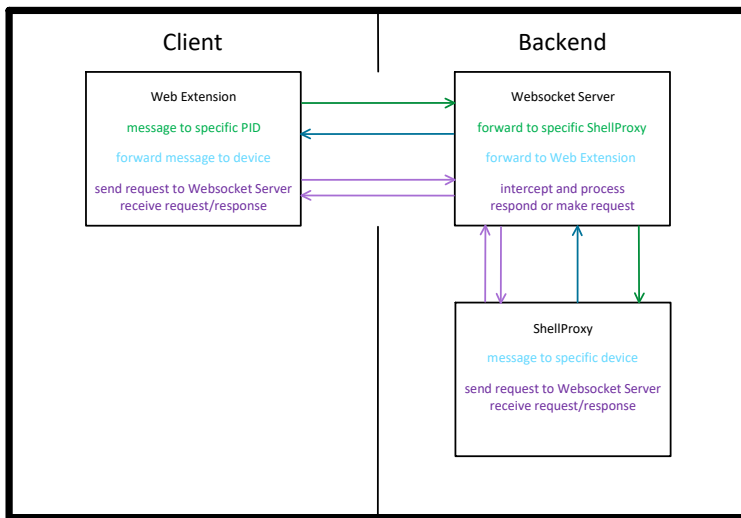
- Connect message**  
send from Web Extension or ShellProxy to inform Websocket Server of new peer
- **when send from Web Extension to Websocket Server**
    - includes id=0 as unique identifier of Client
  - **when send from ShellProxy to Websocket Server**
    - includes id=PID as unique identifier of ShellProxy

Websocket Server maps id to websocket and responds with acknowledgment

**Disconnect message**  
can be send from anywhere to terminate a connection  
is send from Websocket Server when receiving a non (dis-)connect message from an unknown id  
no acknowledgment needed

| Connect (*->WSS)                                      | Acknowledgment (WSS->*)                | Disconnect (*->*)                     |
|-------------------------------------------------------|----------------------------------------|---------------------------------------|
| <pre>{   "type": "connect",   "from": 0   PID }</pre> | <pre>{   "type": "connect ACK" }</pre> | <pre>{   "type": "disconnect" }</pre> |

## Websocket Server routing



After peers have connected, they are able to send messages

Most messages contain a "from" field

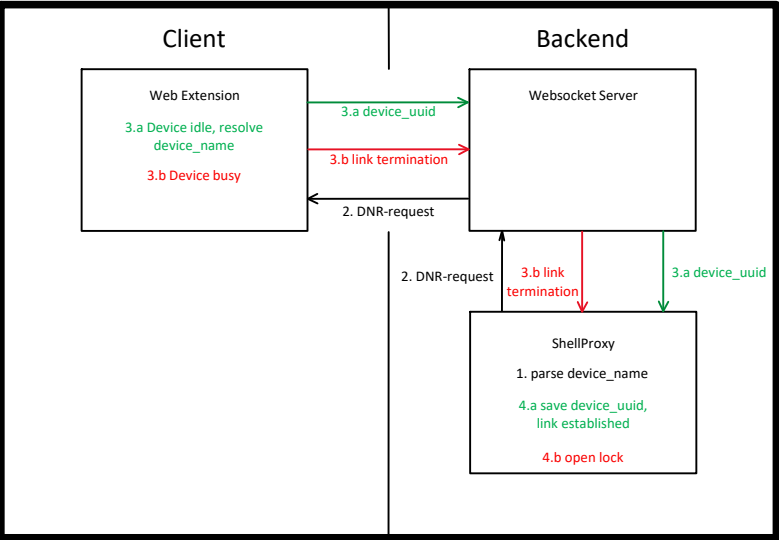
Some messages contain a "to" field

Websocket Server forwards messages with "to"

Websocket Server intercepts messages without "to"

- **Message from Client to ShellProxy**
  - from=device\_uuid, to=PID, Websocket Server forwards message to ShellProxy with PID
- **Message from ShellProxy to Device**
  - from=PID, to=device\_uuid, Websocket Server forwards message to Client with ID=0
  - (Client has to forward according to device\_uuid)
- **Message to Websocket Server**
  - from=PID or from=device\_uuid, is processed by Websocket Server
  - is able to respond or send own requests

Link Establishment from ShellProxy



DNR-request (B->C)    DNR-response (C->B)

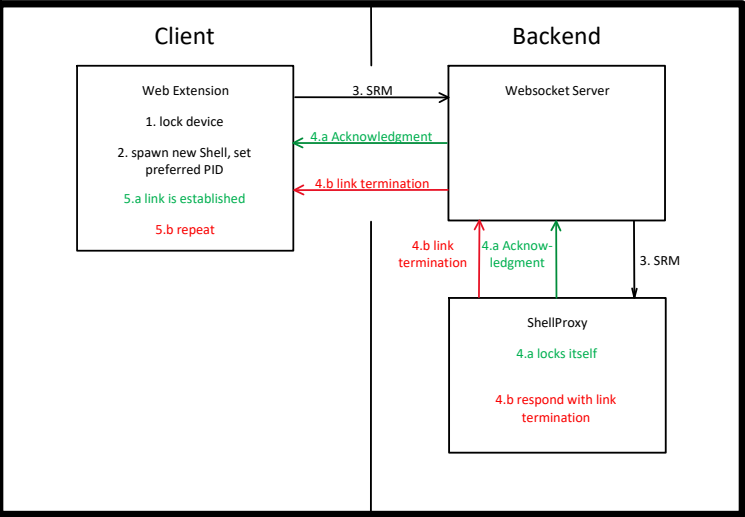
```
{
  "type": "DNR",
  "origin": "B",
  "from": "PID",
  "to": "device_name",
}

{
  "type": "DNR ACK",
  "origin": "C",
  "from": "device_uuid",
  "to": "PID",
}
```

**Link establishment from ShellProxy**  
executed when User accesses a Device from a Shell directly

- ShellProxy intercepts command and parses device\_name
- ShellProxy sends DNR-request (Device Name Resolution) to Client and locks itself
- Client responds to ShellProxy with one of the following
  - device\_uuid when requested device is idle (locks device)
  - link termination when device is busy/locked
- ShellProxy performs one of the following
  - the link is established when receiving a device\_uuid
  - opens lock when receiving link termination

Link Establishment from Web Extension



- Link establishment from Web Extension**  
executed when User presses a Button in the Web Extension  
Corresponding Device has either a cached/preferred PID or not
1. Device locks itself
  2. (if no cached PID present) spawn new Shell, set preferred PID to new Shell PID
  3. Device sends SRM (Shell Request Message) to preferred PID
  4. ShellProxy responds to Device
    - a. if ShellProxy is in non user mode and idle, respond with acknowledgment (locks shell)
    - b. if not, respond with link termination
  5. Device receives message
    - a. if acknowledgment, link is established
    - b. else repeat

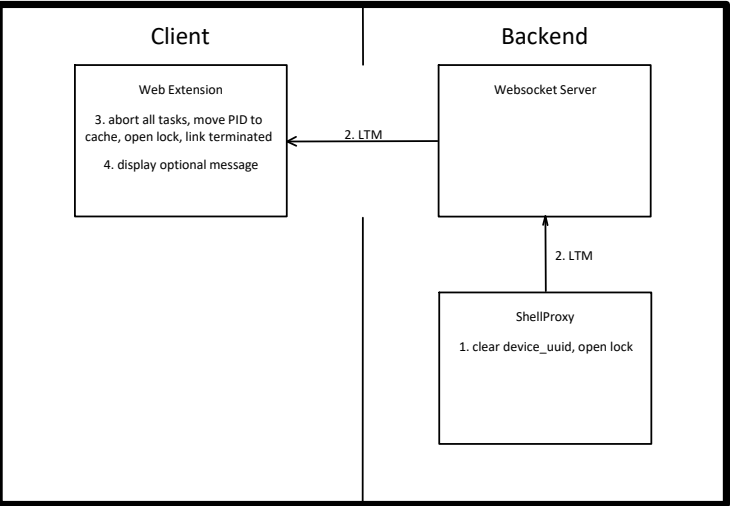
SRM (C->B)

SRM response (B->C)

```
{
  "type": "SRM",
  "origin": "C",
  "from": "device_uuid",
  "to": "PID",
}
```

```
{
  "type": "SRM ACK",
  "origin": "B",
  "from": "PID",
  "to": "device_uuid",
}
```

Link Termination from ShellProxy



Link termination from ShellProxy

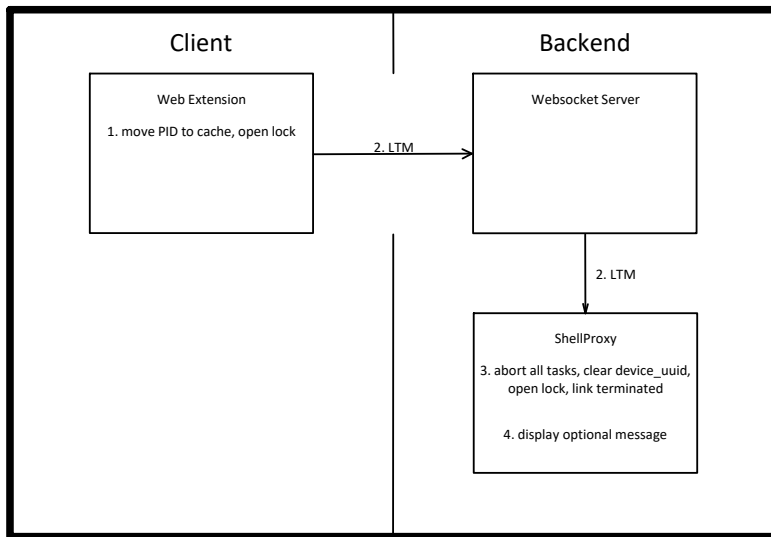
executed when:

- ShellProxy is linked and receives message from unknown device\_uuid
- ShellProxy encounters fatal error
- User interrupts execution

1. ShellProxy clears device\_uuid, opens lock
2. ShellProxy sends LTM (Link Termination Message), as (success or) error, with optional message
3. Device receives message, aborts all tasks, moves PID to cache, opens lock, link is terminated
4. optional message is displayed

```
{
  "type": "LTM",
  "class": "error",
  "msg": msg?,
  "origin": "B",
  "from": PID,
  "to": device_uuid
}
```

## Link Termination from Web Extension



### Link termination from Web Extension

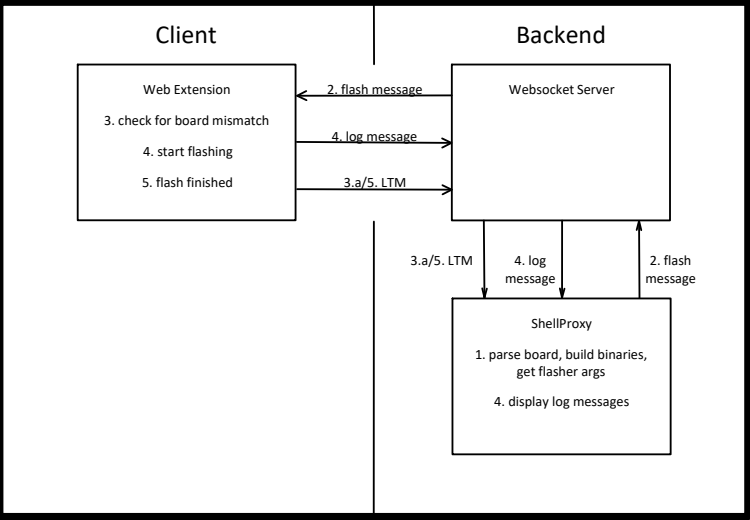
executed when:

- Device is linked and receives message from unknown PID
- Device encounters fatal error
- User interrupts execution
- Task execution is finished

1. move PID to cache, open lock
2. send LTM (Link Termination Message), as success or error, with optional message
3. ShellProxy receives message, aborts all tasks, clears device\_uuid, opens lock, link is terminated
4. optional message is displayed

```
{
  "type": "LTM",
  "class": "success" | "error",
  "msg": msg?,
  "origin": "C",
  "from": device_uuid,
  "to": PID
}
```

Flash from ShellProxy



Flash from ShellProxy

Executed when:

- User inputs make flash in Shell directly with port=device\_name and board=device\_board
- On error or interrupt anywhere at anytime:
- Error LTM is send

0. establish link from ShellProxy
1. parse board, build binaries, get flasher args
2. send flash message to device with board, binaries and flasher args
3. Device checks if correct board is used
  - a. sends error link termination if mismatched
4. device starts flashing, sends log messages to display in ShellProxy
5. On finish, device sends success link termination

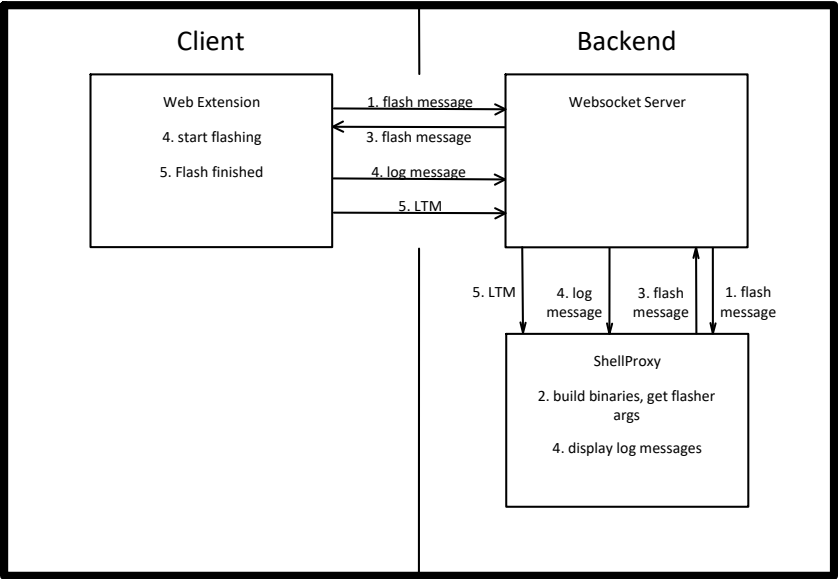
Flash (B->C)

```
{
  "type": "flash",
  "board": board
  "binaries": {
    ...address: binary
  },
  "args": argsstr,
  "origin": "B",
  "from": PID,
  "to": device_uuid
}
```

Log (C->B)

```
{
  "type": "log",
  "class": "log" | "error",
  "msg": msg,
  "origin": "C",
  "from": device_uuid,
  "to": PID
}
```

Flash from Web Extension



Flash from Web Extension

Executed when:

- User presses flash button in Web Extension, has specified board

On error or interrupt anywhere at anytime:

- Error LTM is send

0. establish link from Web Extension
1. send flash message to ShellProxy with board
2. build binaries, get flasher args according to board
3. send flash message to device with binaries, flasher args
4. device starts flashing, sends log messages to display in ShellProxy
5. On finish, device sends success link termination

Flash Request (C->B)

```
{
  "type": "flash request",
  "board": board,
  "origin": "C",
  "from": device_uuid,
  "to": PID
}
```

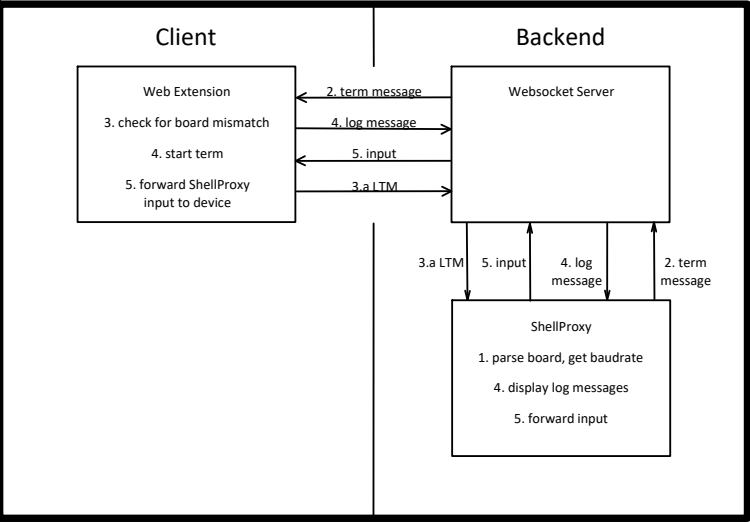
Flash (B->C)

```
{
  "type": "flash",
  "binaries": {
    ...address: binary
  },
  "args": argsstr,
  "origin": "B",
  "from": PID,
  "to": device_uuid
}
```

Log (C->B)

```
{
  "type": "log",
  "class": "log" | "error",
  "msg": msg,
  "origin": "C",
  "from": device_uuid,
  "to": PID
}
```

Term from ShellProxy



Flash from ShellProxy

Executed when:

- User inputs make term in Shell directly with port=device\_name and board=device\_board

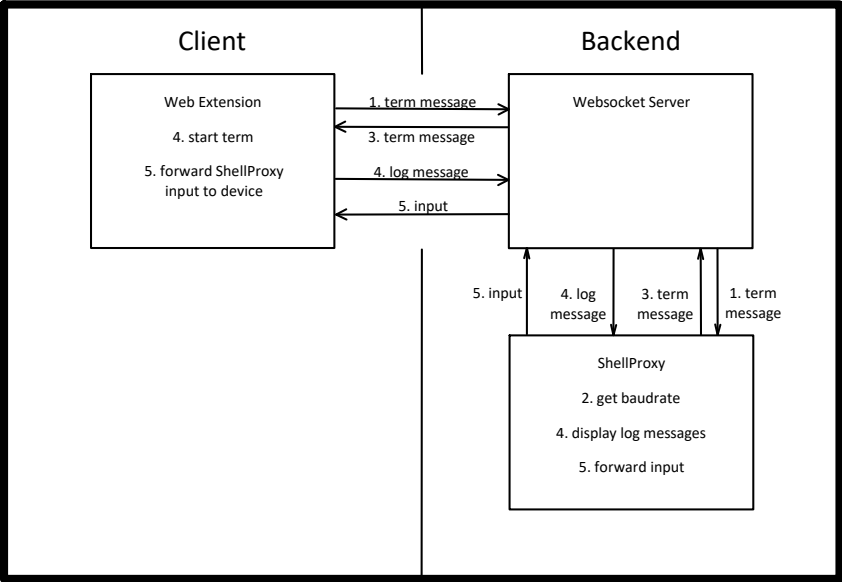
On error or interrupt anywhere at anytime:

- Error LTM is send

0. establish link from ShellProxy
1. parse board, get baudrate
2. send term message to device with board and baudrate
3. Device checks if correct board is used
  - a. sends error link termination if mismatched
4. device starts term, sends log messages to display in ShellProxy
5. ShellProxy forwards inputs to device

| Term (B->C)                                                                                                                    | Log (C->B)                                                                                                                    | Input (B->C)                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <pre>{   "type": "term",   "board": board,   "baudrate": baudrate,   "origin": "B",   "from": PID,   "to": device_uuid }</pre> | <pre>{   "type": "log",   "class": "log"   "error",   "msg": msg,   "origin": "C",   "from": device_uuid,   "to": PID }</pre> | <pre>{   "type": "input",   "input": input,   "origin": "B",   "from": PID,   "to": device_uuid }</pre> |

Term from Web Extension



Flash from Web Extension

Executed when:

- User presses term button in Web Extension, has specified board

On error or interrupt anywhere at anytime:

- Error LTM is send

0. establish link from Web Extension
1. send term message to ShellProxy with board
2. get baudrate according to board
3. send term message to device baudrate
4. device starts term, sends log messages to display in ShellProxy
5. ShellProxy forwards inputs to device

| Term Request (C->B)                                                                                            | Term (B->C)                                                                                                  | Log (C->B)                                                                                                                    | Input (B->C)                                                                                            |
|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <pre>{   "type": "term request",   "board": board,   "origin": "C",   "from": device_uuid,   "to": PID }</pre> | <pre>{   "type": "term",   "baudrate": baudrate,   "origin": "B",   "from": PID,   "to": device_uuid }</pre> | <pre>{   "type": "log",   "class": "log"   "error",   "msg": msg,   "origin": "C",   "from": device_uuid,   "to": PID }</pre> | <pre>{   "type": "input",   "input": input,   "origin": "B",   "from": PID,   "to": device_uuid }</pre> |